

AI reinvention made real | Accenture Innovation Challenge 2026

KPI Intelligence-to-Action Engine

Round 2 — Detailed Business Proposal

Track 3 : BusinessIntelligence.ai

Team Stack Overflowed — IIT Madras

Team details

Member	College	Stream	Graduation
Thirukailash K A	IIT Madras	Mechanical Engineering	2028
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Team name: Stack Overflowed

Describe the problem statement

200 words

A dashboard can show that revenue dropped eight percent in a region, but it usually does not explain why it happened or what the business should do next. Finding the reason often requires an analyst to manually check sales, inventory, pricing, customer feedback, and other data from different systems. This can take days, and by the time the cause is found, it may already be too late to act.

The main problem is that business data is spread across different sources. Numbers like revenue and inventory are stored as structured data, while useful context is hidden in support tickets, emails, and customer reviews. There is also no easy way to tell whether a change in a metric is a real business problem or just normal day-to-day fluctuation.

Because of this, businesses can either react too late or focus on the wrong problem. What they need is a system that can automatically connect different types of data, identify meaningful changes, find their likely causes, and suggest what action to take.

Proposed solution

200 words

Our proposed solution treats every KPI's history as a baseline, using a seasonally adjusted rolling average to define what “normal” looks like for that metric. This means any new value gets checked against an expected range instead of a fixed rule, so everyday fluctuation doesn't get mistaken for a real problem.

When a value falls outside that range, the engine uses the details of that moment — the metric, the region, the time window — as an anchor to investigate. It runs targeted SQL queries against structured data like inventory and pricing for that same time and region, and separately searches unstructured sources like support tickets and emails by traversing a knowledge graph of business entities (regions, warehouses, products), so it can surface a support ticket connected to the anomaly even if the wording never mentions the region or metric directly.

Every piece of evidence is checked before it is trusted: whether it happened before the anomaly, since a cause must precede its effect, and whether it is genuinely relevant, not just structurally nearby in the graph. An AI reasoning step then combines whatever survives into a ranked list of likely causes with confidence levels and a recommended action; if nothing survives with enough confidence, the system reports the cause as unclear instead of guessing.

Solution design : deterministic first, language model last

- Every quantitative claim is computed by deterministic analytics — SQL, statistics, business rules. The language model only words the narrative, and every response records which parts were computed and which were phrased.
- Detection uses two independent signals: a rolling, seasonally adjusted z-score against an eight-period baseline, and an evidence-driven signal that lets support tickets and reviews raise a movement on their own. A period flagged by neither is never admitted.
- Driver ranking uses a deterministic Price-Volume-Mix decomposition for revenue, scoped to the affected item and region so the parts reconcile to the movement exactly.
- Evidence combines structured supply and marketing rows with an anomaly-centric knowledge graph; every item is checked for temporal precedence and genuine relevance before it is trusted.
- A deterministic abstention gate withholds a cause on low confidence, on contradiction, or on a material movement with no isolable driver — no model call.
- A semantic contract fixes every KPI formula, threshold, lineage and access rule; role-based masking is enforced on the server before any response leaves the API.

Target users

Persona	Objective	Decision rights	What the engine delivers
VP of Retail Sales (executive)	Maximise regional revenue and gross margin	Authorise pricing promotions; reallocate marketing budget; initiate supplier renegotiation	250-word executive summary; price / volume / mix framing; one high-level action with a named owner and a monitoring plan
Regional Supply Chain Planner (analyst)	Optimal inventory turnover; zero stockouts	Trigger supplier reorders; approve inter-warehouse transfers; flag lead-time violations	400-word operational report; SKU, warehouse and carrier detail; data-freshness notes; a quantitative reorder action
Data Governance and Compliance	Verify masking; audit decisions	Full read access, granted through the entitlements model	Unredacted ground truth for comparison; abstention and data-quality caveats stated explicitly

Business case and impact

- Analyst time recovered : a material movement that takes three to four hours to investigate across systems is returned as a cited first draft in seconds.
- Faster corrective action : reducing mean time to explanation from about five business days to under one day recovers a large share of the revenue that leaks during the delay.
- Fewer poor decisions : the abstention gate prevents about 77 percent of movements from producing a confident and possibly incorrect recommendation.
- Inference cost avoided : the live path makes no model calls — cost scales with the number of questions asked, not with the number of dashboards rendered.

\$0 per dashboard load 0 model calls on the live path 54 automated tests passing 0 reconciliation mismatches

Evidence and validation

57

material movements detected and
prioritised

44 / 57

held at abstained — figures
returned, cause declined

41 / 8 / 7 / 1

statistical / evidence-driven /
hybrid / sparse mix

8,738 / 40,380

evidence-graph nodes / edges, 0
sign mismatches

27,409

sales rows; the revenue identity
holds on every one

7.44 s / \$0.00

full deterministic seed run,
zero model cost

Every figure is reproducible from the committed database and the 54-test suite.

Phased roadmap

Phase	Scope	Duration	Exit criteria
P0 Prototype (this submission)	Three KPIs; three structured sources plus unstructured text; three personas; full detection, decomposition, evidence graph, abstention and grounded assistant; 54 tests	Complete	Every Round 2 minimum expectation demonstrated on illustrative data
P1 Pilot (single domain)	Connect one production warehouse; replace synthetic feeds with real ones; map roles to single sign-on; add an analyst correction interface that writes back to the contract	~1 quarter	Analyst agreement on ranked drivers at or above 80 percent; 95th-percentile analytics latency below two seconds
P2 Breadth	15 to 30 KPI slices; add forecasting for expected-range bands and causal inference as a third driver method; add proactive alerting to messaging and email	~2 quarters	Drift monitors in production; the feedback loop closes nightly; false-positive alert rate below 10 percent
P3 Scale and governance	Multiple domains on one contract; per-geography policy packs for entitlements and retention; full audit export; a model router meeting cost and latency objectives	~2 quarters	Central governance sign-off; onboarding a new KPI is a contract edit, not a code change

Key risks and mitigations

Risk	Mitigation
The language model fabricates a numerical value	The model performs no computation; it receives only pre-computed, pre-masked evidence and is instructed to treat it as the sole quantitative source. The split is reported on every turn.
Over-flagging leads to alert fatigue	A z-score admission gate, a severity ladder, and materiality ordering. The abstention gate prevents about 77 percent of movements from producing confident output.
Under-flagging leads to a missed movement	Two independent detection signals. A period flagged by neither is never admitted, but either signal alone is sufficient to admit it.
Contradictory or stale source data	A reconciliation layer, plus abstention on contradiction and on a genuinely missing feed. Data freshness is reported in telemetry.
A role sees restricted data (entitlement leak)	Masking is enforced server-side. Ten automated tests assert removal from the payload, including an identifier embedded in a compound key and dollar figures narrated in free text.
Cost or latency growth under load	The live path makes no model calls. The assistant is opt-in, one call per question, free-tier by default, with every token measured.



Video

Prototype walkthrough : add public link (YouTube or Google Drive) before submission

The walkthrough covers, in order: scenario load; the revenue trajectory chart with every series anomaly annotated; the evidence trail and the anomaly-centric knowledge graph; a live role switch that visibly changes field masking; the conversational assistant (a grounded answer, a clarification that makes no model call, and an abstention); approving and assigning an action with an audit identifier; and the runtime telemetry panel.

Thank you

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